

Outlier Aware Open World Test-Time Adaptation

Pattern Recognition | CSE 4835 | Final Presentation

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Motivation

- **Distribution shift:** test data differs from training data
- Accuracy drops sharply. Retraining is usually impossible.
- **Test-Time Adaptation:** adapt a pretrained model online, using only the unlabeled test stream
- No source data. No target labels. Model access only.

Open-world TTA: stay accurate on known classes, and reject unknown ones.

Foundations

BUILT ON



Tent

ICLR'21



EATA

ICML'22



SAR

ICLR'23

THE GAP

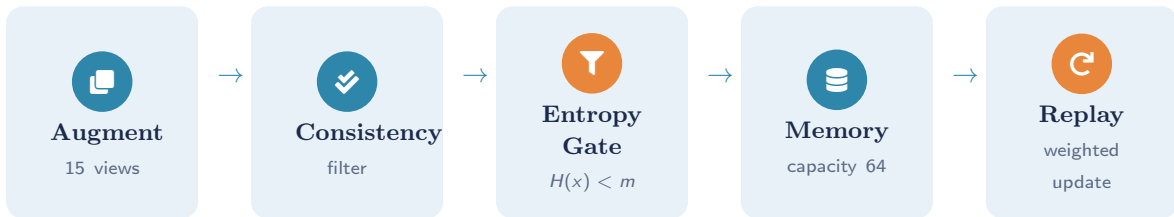
- Most TTA assumes a **closed world**
- Real streams contain **unknown-class outliers**
- Joint recognition and rejection: under-explored

Our Direction

- We build on **STAMP** (Yu et al., ECCV 2024)
- Its detector doubles as the adaptation loss: one signal, two jobs

Research question: can we replace just the detector, and leave the rest of STAMP untouched?

STAMP: The Pipeline



$$\mathcal{L}_{\text{STAMP}} = \frac{1}{|\mathcal{M}|} \sum_{x_i \in \mathcal{M}} \tilde{w}_i H(f_{\theta}(x_i)), \quad \tilde{w}_i = \frac{|\mathcal{M}| w_i}{\sum_j w_j}$$

- Updates only normalization parameters
- **The catch:** entropy decides admission *and* the loss

Setup

- **ID samples:** classify correctly
- **OOD samples:** reject, never adapt on them
- No test labels used, ever
- CIFAR-10-C / 100-C, severity 5, 15 corruptions
- ResNet-18, batch 64

Metrics: Accuracy, AUC, and

$$H\text{-score} = \frac{2 \text{Acc} \cdot \text{AUC}}{\text{Acc} + \text{AUC}}$$

Change only the score. Hold the rest of STAMP fixed.

Roadmap



R1

KNN distance



R2

Energy score



R3

Adaptive
threshold



R4

Loss fix

- R1, R2: change **what** scores a sample
- R3: change **how** the score becomes a decision
- R4: fixes a bug in the replay loss

Round 1: KNN Feature Distance

$$s_{\text{KNN}}(x) = 1 - \frac{1}{k} \sum_{j \in \mathcal{N}_k(x)} \hat{h}(x)^\top \hat{r}_j$$

- Distance to a bank of clean source features
- $k=10$, up to 2,000 references

RESULT

- Accuracy up: **+21.3 pts**
- AUC down: **-9.4 / -11.0 pts**
- Cause: corruption moves the reference bank away from clean ID features

Round 2: Logit Energy Score

$$E_T(x) = -T \log \sum_{c=1}^C \exp(z_c(x)/T)$$

- No reference bank, no extra forward pass
- Threshold: 80th percentile of the last 1,000 scores

RESULT

- CIFAR-100-C: strongest result yet (**71.33 H**)
- CIFAR-10-C: AUC drops to **54.33**
- Not dataset-independent

Round 3: Adaptive Threshold

$$\alpha_{t+1} = \text{clip}_{[0,1]} (\alpha_t + \gamma(\rho - r_t))$$

- Target rejection rate $\rho = 0.20$
- Cutoff adjusts automatically from the rolling score buffer

RESULT

- Matches, but does not improve, the fixed-threshold energy score

Round 4: The Bug

 Before

$$\mathcal{L}_{\text{old}} = \frac{64}{N} \mathcal{L}$$

 After

$$\mathcal{L} = \frac{1}{N} \sum_i \tilde{w}_i H_i$$

- Memory starts partially filled ($N < 64$)
- Gradients were inflated up to **64**×
- Fixed variant: **STAMP+BSN**

RESULT

- Best on both benchmarks: **81.19 H** (C10), **72.89 H** (C100)

Full Results

Dataset	Method	Acc.	AUC	H
CIFAR-10-C	Source	57.34	70.39	62.32
	Norm-test	72.95	68.51	70.58
	KNN (R1)	78.64	59.10	63.61
	Energy (R2)	78.99	54.33	63.80
	Energy+ACI (R3)	78.98	54.08	63.58
	STAMP+BSN (R4)	78.95	83.96	81.19
CIFAR-100-C	Source	35.83	43.13	38.01
	Norm-test	45.90	80.91	58.49
	KNN (R1)	45.73	69.93	54.15
	Energy (R2)	57.14	95.12	71.33
	Energy+ACI (R3)	57.04	95.08	71.24
	STAMP+BSN (R4)	58.02	98.29	72.89

Key Takeaways



KNN: better accuracy, worse detection



Energy: strong on CIFAR-100-C, dataset-dependent



Loss fix: best on both benchmarks



Calibration: no improvement over a fixed threshold

Conclusion

- No single score is dataset-independent
- The biggest gain came from a **correctness fix**, not a new score
- STAMP+BSN: best H-score on both benchmarks

Future Work



Multi-Seed Evaluation

Uncertainty across random seeds



Combine Best of Each Round

Energy + BSN + ACI together



Per-Corruption Diagnostics

ID/OOD rates per corruption



Warm-Up Ablation

Confirm why STAMP+BSN stabilizes C10-C

References

THIS WORK BUILDS ON

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FOUNDATIONAL TTA

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Thank You

Questions & Discussion